

Woon Tian Loong Terence

Experienced Front-End engineer open to develop in other areas of Software Engineering. Learning [Elixir](#) on the side. Currently in Ontario, Canada.

RELEVANT EXPERIENCE

Technology Services Group Pte Ltd (TSGS) — Senior Frontend Developer

June 2024 - December 2024

AI products that help customers trade stocks and options with advice of Large Language Models (LLM) and human curated strategies
Tech includes Nuxt3, Vue.js, ReadableStream, SvelteKit, Svelte 4, Kysely, psql, React, Redux Saga, shadcn(-vue, -svelte), tailwindcss, vite, HighCharts

[PilotAI](#) - Web app that recommends investment portfolios to customers and advises them on whether to hold, sell, or buy individual stocks according to their investment preferences. It also includes a chat interface for users to ask questions about all things investment related, and about their portfolio. It interacts with Tradier's API for (paper) trading of stocks.

I implemented the Trading History feature with filters and sorting so that users can search through their trading history more easily. I also implemented charts within the chat feature of the app. I did many UI improvements such as enlarging the chat area, standardizing the colors, responsive layout, etc.

[TradeGPT](#) - Web app that allows users to chat with AI to get recommendations on how to trade. Gives the users the ability to paper-trade options.

I implemented the feature that allows for users to have multiple chat sessions at once where previously, the chat session would get cancelled if the user switches the chat sessions.

I implemented the detection of the stream's response for Latex syntax so that the UI can render latex.

I also implemented the loaders so that the user can be notified on which tools are working on the query. This was done by looping over the stream response, extracting the keys based on certain keywords, and saving them in a dictionary, of which state needs to be updated when the stream response indicates that the tool has completed running.

[Orion](#) - Internal tool for the company that allows different users (based on their role) to manage user roles, edit and publish release notes, get the latest news summary based on the date and user input, and finally an interface to interact with tables for the financial analyst to create, read, update values and to insert stocks that are white/black listed to the portfolio strategy.

This fullstack project was solely implemented by me including the design and requirements, according to what features I believed were most beneficial to the users. I added a ton of quality of life features such as the ability to use a previously published release note as the template, to have an expandable view for reading the news summary, or to have the date automatically be of the previous date if the current time is before market hours which required working across different time zones.

On the backend, I added backend validation to ensure that the backend was secure.

Besides, I included DB migration which I considered to be good practice.

The tech for this project is SvelteKit (Svelte 4) Shadcn-svelte, Kysely, TypeScript

(437) 557 0492

woontlt@gmail.com

SKILLS

HTML, JavaScript, Typescript, CSS, Svelte, SvelteKit, Elixir, Vue.js, React.js React Native, Ember.js, Swift, Rust, Objective-C, Python3, Git, Unit testing

LANGUAGES

English, Mandarin, JavaScript, Typescript, HTML, CSS, Swift, Rust, Objective-C

E Savvy Pte Ltd — Mobile App Development Engineer

November 2023 - March 2024 (Project was terminated)

Forex and Cryptocurrency Trading platform.

Tech includes UniApp, Vue.js, Websocket, eChart, Python3, Twisted Networking Engine, Redis, MySQL, TCP sockets, Protobufs

OTFX - App for both iOS and Android for Forex and Cryptocurrency.

I implemented live ticker data using websocket to update the K line chart in real time, as well as other UI components typical of apps

Python Server - Backend for connecting to cTrader Trading platform.

I wrote this from scratch to connect to the Manager APIs of cTrader. The connection is done over TCP sockets using Protobufs as the data format. This server creates accounts, gets trader's data, and forex data.

Python Websocket Server - Websocket backend that fetches historical and live forex and ticker data from Redis and sends data to the app clients.

OLEA — Software Engineer

September 2022 - June 2023

A trade finance origination and distribution platform.

Tech includes Node.js & Serverless (Backend), Vue-3 (composition api), Quasar, Typescript

Internal tooling for managing settlements of financing.

Minichain - A prototype blockchain app for investors to see the transactions and projections of their investments with the company. The goal is to replace institutional audits as the basis of trust that their investments are secure.

I improved the UI, implemented "time-travel" snapshots of the projections of the investor, did the hashing of the data using SHA-256, refactored the "datetime"s to use Js-Joda for better accuracy, wrote unit tests, made the UI to display money movement events with pagination.

Asset Booking and Settlements System - A rethinking of the original internal tools with a simpler, more flexible record input system that also caters for the edge cases that the current system is unable to handle.

I integrated what exists outside of a system into an existing system using an iframe, gave it a UI makeover while preserving the UI in the existing system.

- added quality of life improvements to the UX with table filters, transitions and animations.
- did form validation for their input data.
- mentored junior engineers who are not trained in JS/TS
- implemented db migration in post-deploy in a serverless lambda environment
- retrieved bank account details from the db of the existing system while carefully considering the privacy of the account holders.

EVYD Research Private Limited — Front-End Engineer

November 2020 - September 2022

A Health Tech Company providing various solutions to the healthcare industry ranging from data to AI analysis, and healthcare app services.

Front-End Tech in this company is largely Vue.js (2.x) and Element-UI.

Doctor - Patient Management Web App - For clinics to manage the queue of their patients. Features include display of schedule, registration of staff

and patients, epidemic warning and notification.

Among other ordinary features, I implemented the surveillance notification feature which shows a toast message if a professional would sound an epidemic alert. Features include snoozing the alert for a day when the user decides.

Research Data Uploading Web App - For researchers to upload data and to display a high level overview of that data. Features include displaying errors in the data, ignoring invalid data for their analysis, input validation, user authorization etc.

I implemented the error reporting screens. Features are showing errors in a table with table row merging, error logs.

I implemented a data validation rules screen that allow users to set rules on certain kinds of data such as text format, number format, names format, drag and drop to choose specific fields, validation of the rules so that they are coherent

Medical Procurement Web App - Display of drug usage and inventory, with charts to show usage and forecast trends. eCharts.js was used.

I develop the entire front end of this app. Features include tables, data validation, calling backend services, displaying of inventory predictions, user management etc.

Aleph Labs — Senior Front-End Developer

October 2016 - November 2020

Vendor to various Major Banks in Singapore

Standard Chartered Bank (SCB)

UI / UX Uplift — Used an in-house (Aleph) tool using Grunt.js and jQuery to automatically change the styling of the Bank's website.

I used the tool and match css selectors to override the styles of the existing website.

GBPS+ — Built the Web App for SCB for their internal use when answering phone calls concerning commodities. Features include fetching data from their server and Front-End validation. Tech stack was Ember.js (2.x).

I implemented the feature for retrieving and displaying and updating of data. Features include data validation, multiple email inputs.

United Overseas Bank

UOB Mighty (1.x) App — Hybrid App built using Cordova and React.js. I implemented features including displaying data, handling user inputs and interactions, Front-End validation, state management, calling of backend APIs.

UOB Mighty (1.x) App (Lifestyle) - The part within the app that was written with React Native with Redux. Features included displaying dining and lifestyle deals that the bank provides, with animation. I wrote a limited quasi-html parser as a third-party API provided html strings as response that worked with their website.

UOB Mighty (2.x) App - Rewriting of the App to be fully native. I was in the iOS team. The language used was Swift. I was in charge of a sub-team of 5 people. Did code reviews and gave instructions and tips on improving code quality. I also came up with a mini architecture within the app for state management for data shared between screens, and image caching.

EDUCATION

National University of Singapore — *BSc(Hons) Real Estate*

2009 - 2013

CONTINUOUS SELF-LEARNING

Secrets of the JavaScript Ninja — *John Resig, Bear Bibeault*

The Rust Programming Language —by Steve Klabnik and Carol Nichols, with contributions from the Rust Community

Rust in Action — *T.S. McNamara*

Operating System Concepts (Ninth ed.) — Abraham Silberschatz, Peter Baer Galvin, Greg Gagne

PROJECTS

The Psalter 1912 — *Mobile App*

Live app written in React Native and Redux that displays, search, formats texts, plays music. Had to programmatically reduce file size (and quality) of music files. Texts are in a format that is storable as JSON and able to be parsed with formatting. Workaround peculiar quirks around React Native.

GitHub - <https://github.com/iThinkyouknow>

Elixir - https://exercism.org/profiles/iThinkyouknow/solutions?track_slug=elixir

Covid Bar Chart Race of Countries (Svelte JS) - <https://wry-beam.surge.sh>

Exercises in NodeJS

<https://github.com/iThinkyouknow/node-app>

<https://github.com/iThinkyouknow/node-SIP>

Functional Programming in JavaScript

Writing various scripts to automate various tasks in NodeJS

<https://github.com/iThinkyouknow/psalter1912-2/tree/develop/data/data-scripts>

Data Visualization (D3.js)

Choropleth Map: <https://codepen.io/iThinkyouknow/full/LYNRNOY>

Bar Chart: <https://codepen.io/iThinkyouknow/full/BaKzxaX>

Scatterplot: <https://codepen.io/iThinkyouknow/full/jOqrKjQ>

Heatmap: <https://codepen.io/iThinkyouknow/full/jjXKmGq>

Exercises in Rust (Exercism)

<https://github.com/iThinkyouknow/exercism>

Interesting Coding Problems such as Prime Factors, Pythagorean Triplets, Isograms, ISBN Identifier Verification, etc.

Pong Game in Rust

<https://github.com/iThinkyouknow/rust-pong-self>

Other things I do for fun

https://github.com/iThinkyouknow/for_fun